Exercise Sheet 19

Use $(g \cdot \ell)(X) = \ell(\mathrm{Ad}(g^{-1})X)$ and $\omega_\ell(\xi_X, \xi_Y) = \ell([X, Y])$. Heisenberg coordinates are matrix coordinates, with law

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab').$$

Exercises 19.1–19.5 are core. Exercise 19.6 is an optional further example.

Exercise 19.1 (An affine comparison). The connected affine group has matrices $g(a, b) = \left(\begin{smallmatrix} a & b \\ 0 & 1 \end{smallmatrix}\right)$, $a > 0$. Its algebra has $[X, Y] = Y$ and

$$\mathrm{Ad}(g(a, b))X = X - bY, \qquad \mathrm{Ad}(g(a, b))Y = aY.$$

(a) With $p = \ell(X)$ and $q = \ell(Y)$, compute $g(a, b) \cdot \ell$ and list its orbits.

(b) On either two-dimensional orbit, find ξ_X, ξ_Y and the symplectic form in coordinates p, q.

(c) Compute the lower central series. Does the nilpotent orbit theorem apply?

Exercise 19.2 (Pullback of the orbit form). Fix $\lambda \neq 0$ and $\ell_0 = (p_0, q_0, \lambda)$ for the Heisenberg group. Let $\Phi : H \to \mathcal{O}_{\ell_0}$ be the orbit map. Use the rules for 1-forms, the exterior derivative and pullback stated in Lecture 19.

(a) Show that $\Phi^* \omega_\lambda = \lambda \, da \wedge db$.

(b) Verify that $\theta = dc - a \, db$ is left invariant and that $\Phi^* \omega_\lambda = -\lambda \, d\theta$.

(c) Find a primitive for ω_λ on the plane, that is, a 1-form whose exterior derivative is ω_λ. Does the pullback identity imply that the orbit is three-dimensional?

Exercise 19.3 (A circle centre). Let $\Gamma = \{(0, 0, 2\pi k) : k \in \mathbb{Z}\}$ and $\bar{H} = H/\Gamma$.

(a) For which nonzero λ does the Schrödinger representation

$$\Pi_\lambda(a, b, c)u(t) = e^{i\lambda(c+bt)} u(t + a)$$

descend to $\bar{H}$?

(b) Which point-orbit characters descend? Why is it incorrect to attach an irreducible representation of $\bar{H}$ to every Heisenberg plane without a further condition?

Exercise 19.4 (Moving within a plane). Let $\ell = (p, q, \lambda)$ with $\lambda \neq 0$ and use $\mathfrak{m} = \mathrm{span}\{Y, Z\}$.

(a) Compute χ_ℓ on M and the induced action in the section $s(t) = (t, 0, 0)$.

(b) Set $(T_q \varphi)(t) = \varphi(t - q/\lambda)$. Show directly that $T_q \Psi_{0,\lambda}(g) = \Psi_{q,\lambda}(g) T_q$.

(c) Explain why p does not enter this induced formula and why different (p, q) do not give different irreducibles at fixed nonzero λ.

Exercise 19.5 (Hamiltonian coordinates). On a Heisenberg plane, define the Hamiltonian vector field X_f of a smooth real function f by $\iota_{X_f}\omega_\lambda = df$. This means $\omega_\lambda(X_f, v) = df(v)$ for all tangent vectors v. Define $\{f, h\} = \omega_\lambda(X_f, X_h)$.

(a) Compute X_f and $\{f, h\}$ in coordinates.

(b) Check $\{p, q\} = \lambda$. For $\mu_W(\eta) = \eta(W)$, verify $\{\mu_X, \mu_Y\} = \mu_Z$ and the two central brackets.

Exercise 19.6 (Optional extension: a four-dimensional nilpotent algebra). Let $\mathfrak{g}$ have basis X, Y, Z, W with $[X, Y] = Z$, $[X, Z] = W$, and other basis brackets zero. Let G be its connected simply connected group. Write dual coordinates (p, q, z, r), and restrict attention to $r \neq 0$.

(a) Compute the infinitesimal coadjoint vectors and show that r and $C = 2rq - z^2$ are invariant.

(b) Show that the orbit through a functional with $r \neq 0$ is the surface on which r and C take that functional's values. Show also that (p, z) are coordinates on this surface, with $q = (C + z^2)/(2r)$.

(c) Find its symplectic form and a polarisation.